IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: J26-CSNE-310

1. Topic (12 words max)

Adaptive Cyber-Physical Edge System for Fair Active Power Curtailment in Solargrids

2. Research group the project belongs to

CI - Computing Infrastructure

3. Specialization of the project belongs to

Computer Systems and Network Engineering

4. If a continuation of a previous project:

Project ID	No
Year	No

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature/Gravity & Impact of the Problem:

The rapid global and local adoption of residential rooftop solar systems introduces severe technical challenges to legacy distribution networks. During peak sunlight hours, household solar generation often surpasses local demand, forcing electricity to flow backward into the low-voltage distribution transformer. This phenomenon, known as Reverse Power Flow, triggers dangerous voltage spikes exceeding standard regulatory limits (e.g., 240V). If unmanaged, these spikes cause localized hardware degradation, premature transformer burnouts, and frequent inverter tripping, leading to economic losses for utility providers and consumers.

User Persona:

The future users of this solution include electrical utility operators (such as the Ceylon Electricity Board - CEB) managing grid stability, and residential solar prosumers seeking stable financial returns on their renewable energy investments without facing aggressive, uncoordinated system shutdowns.

Research Cluster & Industry Vertical:

This project aligns directly with the Smart Systems and Cyber-Physical Networks research cluster, targeting the Renewable Energy and Smart Grid Utilities industry vertical. Attainment of Sustainable Development Goals (SDGs): This research directly advances SDG 7 (Affordable and Clean Energy) by enabling high-density solar integration, SDG 9 (Industry, Innovation, and Infrastructure) through the creation of resilient, low-cost smart grid frameworks, and SDG 11 (Sustainable Cities and Communities) by fostering localized sustainable energy production.

References

[1] Ceylon Electricity Board, "Renewable Energy," CEB Knowledge Hub, ceb.lk.

[2] "Distribution Transformer Based Smart Grids with Rooftop Solar: A Case Study for Sri Lanka," IEEE Conference Publication, 2018.

[3] Asian Development Bank, "Rooftop Solar Power Generation Project (RRP SRI 50373-002)," Sector Assessment.

[4] "Enhancing PV Hosting Capacity Using Voltage Control and Employing Dynamic Line Rating," Energies, vol. 15, 2022.

6. Brief description of Existing Research and Systems (300 words max)

Current industry practices rely heavily on Centralized Optimal Power Flow (OPF) systems or rigid, localized communication-free droop settings (e.g., Volt-Var/Volt-Watt controls) pre-programmed into individual solar inverters. Academically, modern research focuses on deploying complex cloud-driven Machine Learning architectures to forecast generation patterns and schedule grid adjustments. From a communications perspective, Software-Defined Networks (SDN) and standard MQTT protocol layers are frequently evaluated to move smart meter telemetry packets under steady network environments.

Gaps/Limitations Identified:

Existing methodologies suffer from four critical limitations:

1. **Unfair Local Enforcement**: Traditional model-free localized active power curtailment (APC) disproportionately penalizes prosumers located at the far end of the feeder lines, forcing them to throttle down more solar than households closer to the transformer.

2. **Static Network Rules**: Standard smart grid communication architectures rely on generic QoS rules that remain static, failing to adapt network security parameters or routing priorities dynamically when physical electrical nodes enter high-voltage crises.

3. **High Operational Latency**: Centralized or offline Hosting Capacity Estimation (HCE) models cannot calculate allowable PV exports fast enough to respond to cloud-induced, sub-second solar drops, demanding heavy communication infrastructure that is economically non-viable for developing regions.

4. **Cloud Cost Inefficiencies**: Heavy cloud-edge frameworks naively stream massive volumes of high-frequency IoT data to central servers, incurring unsustainable cloud storage and bandwidth costs without considering grid state urgency

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressing the Gaps:

This project presents an integrated, low-cost cyber-physical infrastructure designed to manage high-density solar integration safely and fairly without requiring expensive grid infrastructure upgrades. By implementing an optimized Hardware-in-the-Loop (HIL) testing model, we bridge real physical hardware with automated local intelligence to solve the structural gaps of unfair curtailment, static communication protocols, slow cloud computational delays, and cloud data bloat.

System Perspective of the Solution:

- **Input Layer**:
  - Physical: Real-time instantaneous analog voltage (V) and current (I) waveforms captured via non-invasive hardware sensors at a physical lab bench.
  - Virtual: High-fidelity consumption and generation smart meter time-series datasets (Pecan Street Dataport) representing a 50-household neighborhood, combined with live NREL weather API variables (solar irradiation, ambient temperature, cloud cover).
- **Process Layer:**
  - Embedded Node: Applies an on-chip Kalman filter to clean up AC noise and executes an algorithmic fairness curtailment loop.
  - Network Engine: Runs an adaptive MQTT broker that activates lightweight ASCON encryption and upgrades message routing streams to high-priority QoS pathways only when a physical voltage limit is breached.
  - Edge Intelligence Node: Hosts a lightweight Machine Learning model (Random Forest/XGBoost) within an optimized Docker microservice to approximate heavy grid equations and estimate neighborhood hosting capacity.
  - Cloud Analytics Node: Executes a deep LSTM network for 24-hour look-ahead solar forecasting and processes a grid-aware data-tiering compression logic.
- **Output Layer**:
  - ○ Dynamic, localized, and fair active solar throttling control commands sent back to the microcontroller node.
  - ○ Real-time web monitoring dashboards (Grafana/Node-RED) displaying compressed operational insights and macro-level forecasts for utility operators.

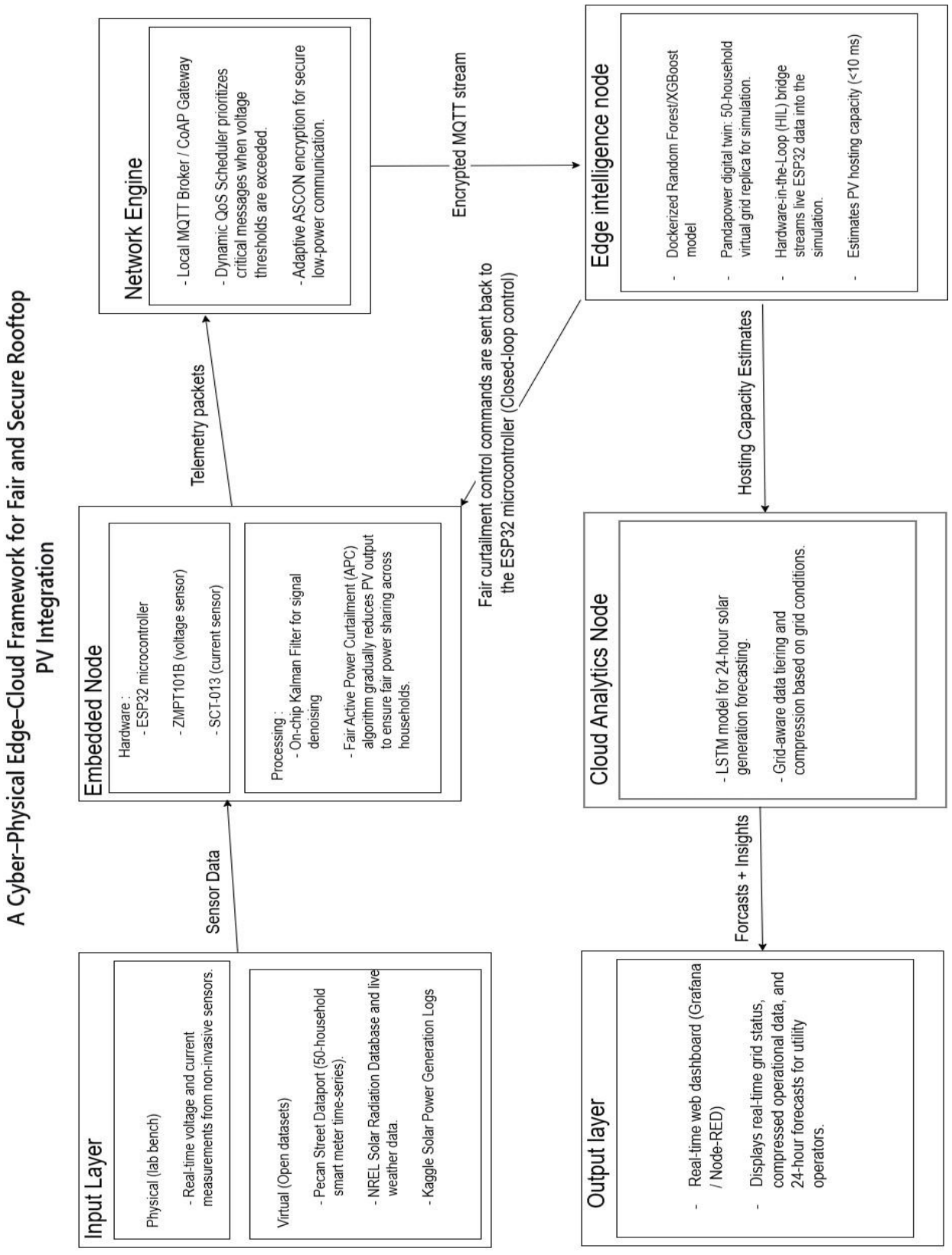


Figure 1: Conceptual architecture of the Context-Aware Edge-Cloud Framework for Dynamic Solar Grid Integration

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain Expertise, Knowledge, & Technical Skills Acquisition:

The project team has structured a collaborative knowledge-acquisition pipeline to gain the necessary domain expertise across electrical systems and computer engineering. Power systems simulation and grid modeling parameters will be acquired through deep study of standard IEEE distribution test feeders and open-source documentation of the Pandapower ecosystem. Advanced technical skills in embedded firmware optimization, time-series data handling, machine learning design, and network protocol engineering will be developed through verified academic pathways, including specialized peer-reviewed literature, cloud architecture documentation (AWS/GCP), and technical handbooks for embedded systems.

Data Collection Methodology:

The data required to train, validate, and evaluate the proposed cyber-physical system is fully available and accessible via open-source, peer-reviewed repositories, eliminating any external dependency or deployment delays:

1. Grid Data: Real-world high-frequency household consumption and solar generation time-series logs are sourced directly from the globally recognized Pecan Street Dataport (Texas, USA).
2. Weather and Yield Data: Macroscopic climate metrics and historical photovoltaic yield patterns are retrieved from the Kaggle Solar Power Generation Logs and the National Renewable Energy Laboratory (NREL) National Solar Radiation Database.

3. Physical Telemetry: Real-time localized inputs are generated programmatically on the physical testbench using calibrated sensor inputs.

Ethical Clearance:

This project does not interact with human subjects, collect personally identifiable information (PII), or modify active public utility infrastructures. Since it relies completely on public open-source datasets and safe, isolated low-voltage laboratory test benches, no ethical clearance is required.

9. Objectives and Novelty

Main Objective
To design, implement, and validate a cyber-physical edge–cloud framework that improves the integration of residential rooftop solar systems into distribution grids by combining fair active power curtailment, adaptive communication, edge-based intelligence, and cloud analytics using a Hardware-in-the-Loop (HIL) environment

Member Name with Registration No	Sub Objective	Tasks	Novelty
Sandes O M S IT 23314924	Design and implement an intelligent house-side edge device that monitors local solar voltage/current and executes power curtailment safely, smoothly, and autonomously, ensuring reliable operation even when central coordination is delayed or unavailable.	• Continuously sense voltage and current at the house's solar output using onboard sensors, and compute real-time power. • Transmit sensor readings (voltage, current, power) to the central transformer-level system at regular intervals via the communication network. • Receive and parse curtailment instructions (target power reduction percentage) from the central system, tracking the timing of each instruction received. • Execute received curtailment instructions through a smooth, stepped power ramp-down using PWM-controlled actuation, continuously verifying against live sensor feedback during the ramp. • Monitor for communication timeout; if no valid instruction arrives within a defined window and local readings indicate an unsafe voltage condition, autonomously apply a pre-defined local curtailment rule until central coordination resumes. • Physically actuate the calculated power reduction (whether from normal or fallback mode) using a MOSFET/relay-controlled load, simulating inverter power-limiting behavior. • Build and validate a simulated solar-plus-load testbed, in the absence of a real grid-tied inverter, to test response time, ramping smoothness, and	A hybrid local control mechanism for house-side curtailment actuation that combines smooth, sensor-verified ramped power reduction with autonomous fallback control — enabling the device to safely execute central curtailment commands under normal conditions, while independently maintaining safe voltage levels using only local sensor data when central communication is delayed or unavailable. This ensures both operational smoothness and system resilience at the household level, without dependency on the internal workings of the detection (ML) or communication components.

		fallback correctness.	
Wimalasena D C B V D S IT23150652	To design and develop a reliable communication system that enables real-time data exchange between the embedded monitoring node, edge device, and cloud platform, ensuring that important grid information and control commands are delivered efficiently under different grid conditions.	* Design the communication architecture between the ESP32 smart metering node, edge device, and cloud platform. * Implement a lightweight MQTT and CoAP communication framework for exchanging sensor data, hosting capacity estimates, and power curtailment commands. * Develop a communication priority mechanism that automatically gives higher priority to critical messages, such as voltage violations and curtailment commands, while allowing routine monitoring data to use normal communication priority during stable grid conditions. * Implement lightweight ASCON encryption to protect telemetry data and control messages exchanged between system components. * Evaluate communication performance by measuring latency, packet delivery success, response time, reliability, and the communication overhead introduced by security under different grid operating conditions.	Most existing smart grid communication systems transmit all messages using fixed communication settings regardless of the current condition of the electrical grid. As a result, critical control messages may experience the same communication behaviors as routine monitoring data. This research introduces a grid-aware communication mechanism that automatically prioritizes important messages, such as voltage alerts and active power curtailment commands, while maintaining lightweight security using ASCON encryption. This improves the responsiveness of the overall cyber-physical system without introducing unnecessary communication overhead during normal grid operation.
Wanniarachchi C P IT23184176	Develop an edge intelligence module that estimates local PV hosting capacity combining a Hardware-in-the-Loop (HIL) environment	• Build a digital twin of a residential distribution network, including the distribution transformer, using Pandapower. • Develop a Hardware-in-the-Loop (HIL) environment that integrates five ESP32 smart metering nodes representing residential households with the digital twin, enabling real-	Traditional hosting capacity analysis usually depends on computationally expensive power flow calculations that are performed offline or on centralized servers. These approaches are too slow for real-time decision-making. This research replaces repeated power flow calculations with a

	with a lightweight machine learning model running on a Raspberry Pi edge device.	time interaction between physical hardware and the simulated distribution network. • Deploy a Raspberry Pi as the edge device to receive real-time measurements from the five ESP32 nodes, receive simulated transformer measurements from the digital twin through the HIL environment, combine both data sources, and process them locally for real-time decision-making. • Generate training data under different grid operating conditions using the digital twin and train a lightweight machine learning model, such as Random Forest or XGBoost, to estimate the available PV hosting capacity. • Deploy the trained model on the Raspberry Pi and evaluate its hosting capacity estimation accuracy and execution time using the HIL environment while validating the system with the five physical ESP32 nodes.	lightweight machine learning model running at the edge. By learning from simulation data, the model can estimate the maximum safe PV capacity within milliseconds, enabling faster local decision-making.
Somachandra K P D M IT23199194	To develop an intelligent, grid-aware cloud data platform that reduces cloud storage costs and network bandwidth by adaptively compressing routine solar monitoring data during stable grid conditions while preserving high-resolution data during grid anomalies, integrated with a 24-hour AI-driven solar	* Collect historical solar generation and weather data from publicly available datasets * Train and evaluate an LSTM model to forecast PV generation up to 24 hours ahead. * Design a grid-aware data tiering and compression mechanism that adjusts stored data based on current grid conditions. * Preserve high-resolution operational data during abnormal grid events while reducing storage requirements during stable operating periods. * Develop a real-time monitoring dashboard using Grafana or Node-RED to visualize forecasts,	Many cloud-based monitoring systems continuously store all sensor data at full resolution regardless of whether the electrical grid is operating normally or experiencing disturbances. This results in unnecessary cloud storage and bandwidth usage. This research introduces a grid-aware data management approach that changes how data is stored based on the condition of the grid. Detailed data is preserved during abnormal events for analysis, while routine monitoring data is compressed during normal operation to reduce cloud resource usage without losing important information.

	power forecast and a real-time monitoring dashboard.	operational data, and system status.	

10. This section is to be completed by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Data required for the project is available.

Yes	✓	No	

f) Any other comments:

This is approved to proceed with

	Title	First Name	Last Name	Signature
Supervisor*	Prof	Amadha	Jayalath	
Co-Supervisor*	Ms.	Shashika	Lankeja	15/7/26
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.

CamScanner

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This part is to be filled by the Topic Screening Panel members.

Project ID: J26-CSNE-310

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes (should be followed up by the supervisor)*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Section 1 :- Capturing the waveform is NOT relavent.

Section 2 :-

- Add more ESP32 sensors (e.g. 5)
- Propose a syster which can combine the whole components.
- Discuss with a power engineer on the whole process.

The Review Panel Details

Member's Name	Signature	Date
Nimal Ratnayake		31/07/2026
Harsha Mahaadikara		31/07/2026

*Important:

1. According to the comments given by the panel, make the necessary modifications and get approval from the **Supervisor** or the **Same Panel**.
2. If the project topic is rejected, identify a new topic, and follow the same procedure until the topic is approved by the assessment panel.